Form C

Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Complete the conversion:
5 pounds = _____ ounces

A. 0.3
B. 21
C. 50
D. 80

D (1 lb = 16 oz, so 5 × 16 = 80 oz)

2. Which numbers are equivalent to $4\frac{2}{5}$? Circle all that apply.

A. 4.25
B. 4.4
C. $\frac{22}{5}$
D. $4\frac{4}{10}$
E. $4.\overline{4}$
F. $\frac{5}{22}$

B, C, D (2 ÷ 5 = 0.4; 4 × 5 + 2 = 22, so 22/5; 4/10 = 2/5)

3. Nine of 21 players on a soccer team wore red jerseys. What is the ratio of players who wore red jerseys to players who did not wear red jerseys?

A. 3 : 7
B. 4 : 7
C. 4 : 3
D. 3 : 4

D (21 − 9 = 12 did not; 9 : 12 = 3 : 4)

4. A bag holds only red and blue marbles. The ratio of red marbles to blue marbles is 4 : 1. What is the ratio of blue marbles to total marbles in the bag?

A. 1 : 4
B. 1 : 5
C. 4 : 5
D. 5 : 1

B (total parts = 4 + 1 = 5; blue : total = 1 : 5)

5. A train traveled 210 miles in 3.5 hours. What was its speed in miles per hour?

A. 60 miles per hour
B. 65 miles per hour
C. 70 miles per hour
D. 73.5 miles per hour

A (210 ÷ 3.5 = 60 mph)

6. Which rate should be used to convert $\frac{3 \text{ gallons}}{1 \text{ day}}$ to $\frac{\text{gallons}}{\text{hour}}$?

A. $\frac{24 \text{ hours}}{1 \text{ day}}$
B. $\frac{1 \text{ day}}{24 \text{ hours}}$
C. $\frac{4 \text{ quarts}}{1 \text{ gallon}}$
D. $\frac{1 \text{ gallon}}{4 \text{ quarts}}$

B (day is on the bottom, so put day on top: 3 ÷ 24 = $\frac{1}{8}$ gallon per hour)

7. For numbers 7a – 7d, circle YES or NO to tell whether the rate given is equivalent to a pump moving 900 gallons per hour.

7a. 15 gallons per minute	YES	NO
7b. 54,000 gallons per minute	YES	NO
7c. 21,600 gallons per day	YES	NO
7d. 0.25 gallon per second	YES	NO

7a YES (900 ÷ 60 = 15) · 7b NO (that multiplies by 60) · 7c YES (900 × 24 = 21,600) · 7d YES (900 ÷ 3,600 = 0.25)

8. What is the unit rate for $\frac{\frac{4}{5} \text{ liter}}{2 \text{ minutes}}$?

A. $1\frac{3}{5}$ liters per minute
B. $2\frac{1}{2}$ liters per minute
C. $\frac{4}{5}$ liter per minute
D. $\frac{2}{5}$ liter per minute

D ($\frac{4}{5} \div 2 = \frac{4}{5} \times \frac{1}{2} = \frac{4}{10} = \frac{2}{5}$)

9. Jada hiked $4\frac{2}{3}$ miles in $1\frac{3}{4}$ hours. Assume she hiked the entire time at the same rate. What was her speed in miles per hour?

A. $8\frac{1}{6}$ miles per hour

B. $\frac{3}{8}$ mile per hour

C. $2\frac{2}{3}$ miles per hour

D. $3\frac{1}{2}$ miles per hour

C ($\frac{14}{3} \div \frac{7}{4} = \frac{14}{3} \times \frac{4}{7} = \frac{56}{21} = \frac{8}{3} = 2\frac{2}{3}$ mph)

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Part II – Constructed Response

Complete each conversion.

1. 7 yards = _____ feet

21 feet (7 × 3)

2. 9 centimeters = _____ millimeters

90 millimeters (9 × 10)

3. 3,000 pounds = _____ tons

1.5 tons (3,000 ÷ 2,000)

4. 54 inches = _____ feet

4.5 feet (54 ÷ 12)

Convert each fraction to a decimal.

5. $\frac{9}{20}$

0.45 (9 ÷ 20, or $\frac{45}{100}$)

6. $\frac{5}{6}$

$0.8\overline{3}$ (5 ÷ 6 = 0.8333...)

Convert each decimal to a fraction.

7. 0.8

$\frac{4}{5}$ ($\frac{8}{10} = \frac{4}{5}$)

8. 0.24

$\frac{6}{25}$ ($\frac{24}{100} \div 4 = \frac{6}{25}$)

9. Simplify the ratio 25 : 15.

5 : 3 (divide both by 5)

10. Ten of 25 runners on the cross-country team finished the race in under 30 minutes. Write three different simplified number ratios using this information. Explain the meaning of each ratio using words.

Ratio (numbers):	Ratio (numbers):	Ratio (numbers):
Explanation: 2 : 5 — runners under 30 minutes to all runners (10 : 25)	Explanation: 3 : 5 — runners not under 30 minutes to all runners (15 : 25)	Explanation: 2 : 3 — runners under 30 minutes to runners who were not (10 : 15). The reversed ratios 5 : 2, 5 : 3 and 3 : 2 are also correct if explained.

Find the unit rate.

11. $\frac{72 \text{ feet}}{6 \text{ seconds}}$

12 feet per second (72 ÷ 6)

12. $\frac{\frac{3}{5} \text{ gallon}}{\frac{9}{10} \text{ minute}}$

$\frac{2}{3}$ gallon per minute ($\frac{3}{5} \times \frac{10}{9} = \frac{30}{45} = \frac{2}{3}$)

13. Grace spent $8.75 on 5 notebooks. Omar spent $13.20 on 8 notebooks. Which person paid less per notebook? Show all work necessary to justify your answer.

Grace: $8.75 ÷ 5 = $1.75 per notebook. Omar: $13.20 ÷ 8 = $1.65 per notebook. Omar paid less per notebook.

14. A deer runs at a rate of 30 miles per hour. Write this rate as feet per hour. Show all work necessary to justify your answer.

158,400 feet per hour ($\frac{30 \text{ mi}}{1 \text{ hr}} \times \frac{5{,}280 \text{ ft}}{1 \text{ mi}} = 158{,}400$ ft per hr)

15. A cyclist rides at a rate of 21 kilometers per hour. Write this rate as meters per minute. Show all work necessary to justify your answer.

350 meters per minute ($\frac{21 \text{ km}}{1 \text{ hr}} \times \frac{1{,}000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{60 \text{ min}} = \frac{21{,}000 \text{ m}}{60 \text{ min}} = 350$ m per min)

16. Nina walked $\frac{2}{3}$ mile in $\frac{1}{4}$ hour. Assume she walked the entire time at the same rate. What was her speed in miles per hour? Caleb answered this question, but made a mistake. Explain Caleb's mistake. Then complete the work correctly to answer the question.

I need to find: $\frac{\frac{2}{3} \text{ mile}}{\frac{1}{4} \text{ hour}}$

$\frac{\frac{2}{3} \text{ mile}}{\frac{1}{4} \text{ hour}} = \frac{2}{3} \div \frac{1}{4}$

$= \frac{3}{2} \cdot \frac{1}{4}$

$= \frac{3}{8}$ mile per hour

Mistake: Caleb flipped the first fraction ($\frac{2}{3}$) instead of the divisor ($\frac{1}{4}$). Keep the first fraction and multiply by the reciprocal of the second. Correct: $\frac{2}{3} \times \frac{4}{1} = \frac{8}{3} = 2\frac{2}{3}$ miles per hour.

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Part III – Problem Solving

The science club has 8 seventh graders and 18 eighth graders. They fold paper cranes for a hospital fundraiser, and a sponsor donates money for each crane. Each seventh grader earned $3.00 per crane and folded at a rate of 15 cranes per hour. Each eighth grader earned $2.00 per crane and folded at a rate of 4 cranes every 20 minutes. What is the ratio of the amount of money the seventh graders raised in one hour to the amount of money the eighth graders raised in one hour?

Seventh graders: 8 × 15 cranes × $3 = $360 per hour. Eighth graders: 4 cranes per 20 min = 12 cranes per hour (60 ÷ 20 = 3, 4 × 3 = 12); 18 × 12 cranes × $2 = $432 per hour. Ratio 360 : 432 = 5 : 6 (divide both by 72).